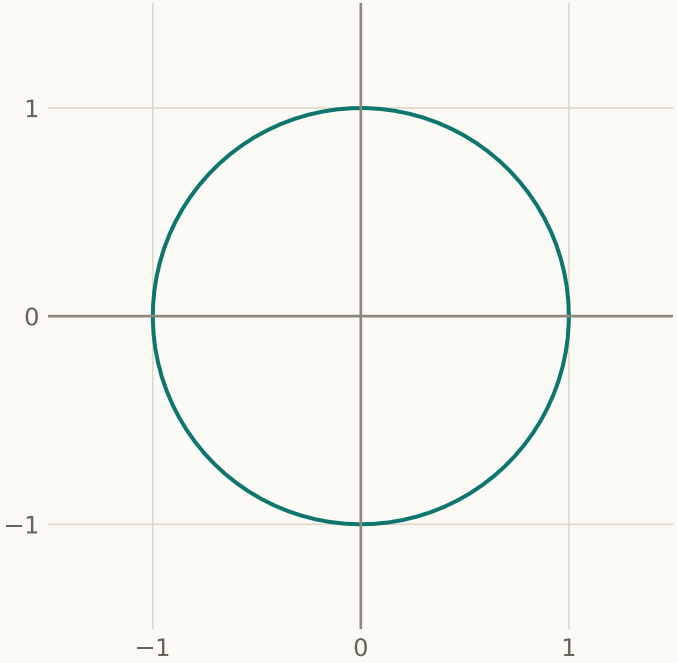
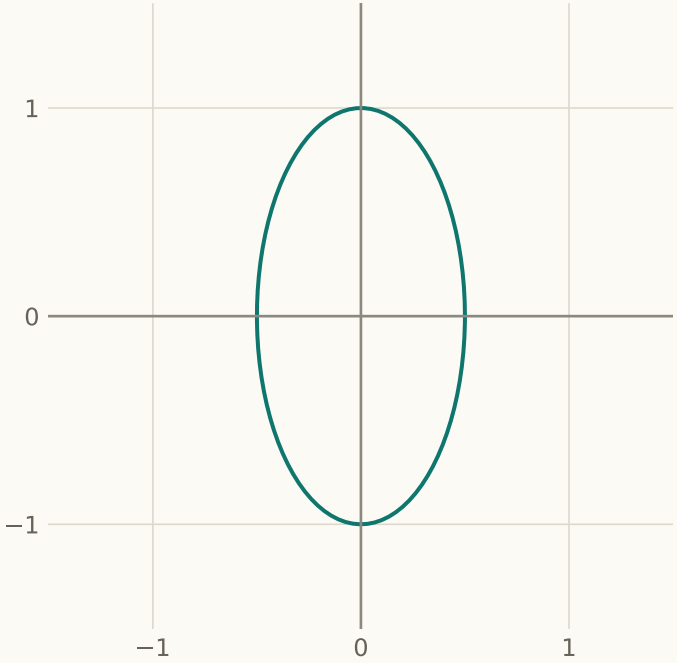
A metric changes which arrows have unit length

g = I: x² + y² = 1



g = diag(4,1): 4x² + y² = 1